

CRT — RESYNCHRONIZATION

fixes dyssynchrony · not sudden death

FUNCTIONAL MR

CARE-HF · CRT
↓ all-cause mortality
NYHA III–IV

MADIT-CRT
+ RAFT
• NYHA I–II
• ↓ HF events

CRITERIA BOARD

CLASS I — STRONGEST EVIDENCE

- LVEF ≤35
- NYHA II–IV
- LBBB · QRS ≥150
- SINUS RHYTHM
- OPTIMAL GDMT ≥3 mo

CLASS II — weaker / selective

- LBBB QRS 120–149
- NON-LBBB QRS ≥150
- AF — only if BiV ≥95%
(often AV node ablation)

QRS DURATION = strongest predictor

AVOID CRT

1 NARROW QRS < 120

2 ECCO-only dyssynchrony < 130

3 ISOLATED RBBB

4 PRESERVED EF

5 AF without biv pacing

WON'T WORK

ECG TICKER TATION

- LBBB · QRS 160 best responder ✓
- RBBB ?
- Narrow QRS ✗

CRT = pump · ICD = anti-death

lower SCD risk
• older / frail / limited life

younger
• ISCHEMIC CM
• higher SCD risk

CRT-P

CRT-D

PREVENT pacing-induced CM

PACING-DEPENDENT + EF ≤35

RV PACING ✗

LEFT WING

DYSSYNCHRONY
↓ systole ↓ efficiency
↓ filling ↑ wall stress
↑ functional MR

DISSONANT

REVERSE REMODELING
↓ MORTALITY · ↓ HOSP · ↑ EF · ↑ EXERCISE

≥150 ms · LBBB

LV WALLS

CRT CENTER STAGE

RV

3-MONTH GDMT WAIT unless complete heart block

FRONT ROW

1. CRT fixes dyssynchrony, not sudden death

WHAT YOU SEE

Title: CRT resynchronization, not sudden death

WHAT IT ENCODES

CRT (cardiac resynchronization therapy) is biventricular pacing — a third lead in the coronary sinus paces the LV lateral wall simultaneously with the RV so the dyssynchronous, LBBB-delayed ventricle contracts in coordination. This improves stroke volume, reduces functional MR and drives reverse remodeling. CRT by itself does NOT prevent sudden cardiac death — that role belongs to the ICD, which is why high-SCD-risk patients get a combined CRT-D device.

BOARD TRAP

WRONG: CRT is implanted to prevent sudden cardiac death. Discriminator: CRT corrects electrical/mechanical dyssynchrony and pump function; only the defibrillator component (ICD/CRT-D) terminates VT/VF. A pacing-only CRT-P does not protect against SCD.

2. Class I — strongest evidence

WHAT YOU SEE

Criteria board Class I list

WHAT IT ENCODES

CRT Class I (all must be present): LVEF $\leq 35\%$ + NYHA class II–IV symptoms + LBBB morphology + QRS ≥ 150 ms + sinus rhythm, on optimal GDMT for ≥ 3 months. This is the ideal-responder bundle (think LBBB + wide QRS + low EF). Landmark trials: COMPANION, CARE-HF (mortality benefit), MADIT-CRT and RAFT (benefit even in mild NYHA I–II disease).

BOARD TRAP

WRONG: CRT is indicated for any wide QRS regardless of morphology. Discriminator: the Class I indication specifically requires LBBB AND QRS ≥ 150 ms; non-LBBB or narrower QRS drops you into weaker Class II categories with far less benefit.

3. LVEF $\leq 35\%$

WHAT YOU SEE

LVEF ≤ 35 criterion

WHAT IT ENCODES

CRT requires a depressed LVEF $\leq 35\%$, measured after ≥ 3 months of optimal guideline-directed medical therapy (the same EF threshold used for primary-prevention ICD). The combination of low EF + wide LBBB defines the substrate that benefits from resynchronization.

BOARD TRAP

WRONG: implant CRT on the first echo before optimizing meds. Discriminator: you must re-measure EF after ≥ 3 months of GDMT, because medical therapy alone often recovers EF above 35% and removes the indication.

4. LBBB + QRS ≥ 150

WHAT YOU SEE

LBBB QRS ≥ 150 line

WHAT IT ENCODES

The best CRT responders are LBBB morphology with QRS ≥ 150 ms — the wide LBBB represents the electrical delay CRT actually corrects. Response is graded: LBBB + QRS ≥ 150 ms gives the largest benefit; LBBB + 120–149 ms is intermediate; non-LBBB benefits little. Women and non-ischemic cardiomyopathy tend to respond best.

BOARD TRAP

WRONG: RBBB responds as well as LBBB to CRT. Discriminator: RBBB does not produce the LV lateral-wall delay that CRT corrects, so RBBB patients derive minimal benefit — RBBB is a poor-responder pattern, not an indication.

5. Sinus rhythm + optimal GDMT ≥ 3 mo

WHAT YOU SEE

Sinus rhythm, optimal GDMT ≥ 3 mo

WHAT IT ENCODES

Class I CRT criteria require sinus rhythm and ≥ 3 months of optimal GDMT (ARNI/ACEi, beta-blocker, MRA, SGLT2i, titrated). Sinus rhythm matters because consistent biventricular capture is needed; in AF you can't reliably achieve it without rate control or AV node ablation.

BOARD TRAP

WRONG: start CRT immediately at diagnosis to 'get ahead' of the disease. Discriminator: GDMT must be optimized for ≥ 3 months first — many patients improve EF and symptoms enough on meds alone that they no longer meet criteria.

6. Class II — weaker / selective

WHAT YOU SEE

Class II box: LBBB 120–149, non-LBBB ≥ 150 , AF if BiV $\geq 95\%$

WHAT IT ENCODES

Weaker (Class IIa/IIb) indications: LBBB with QRS 120–149 ms; non-LBBB morphology only if QRS ≥ 150 ms; and AF patients only if near-100% biventricular pacing can be guaranteed (BiV $\geq 95\%$, often requiring AV node ablation). Benefit shrinks as you move down this list.

BOARD TRAP

WRONG: an AF patient with low EF and wide QRS gets the same CRT benefit as a sinus-rhythm patient. Discriminator: in AF, intrinsic conduction competes with the device, so without AV node ablation BiV pacing falls below the $\sim 95\%$ threshold needed for benefit.

7. QRS duration = strongest predictor

WHAT YOU SEE

QRS duration strongest predictor box

WHAT IT ENCODES

QRS duration is the single strongest predictor of CRT benefit — the wider the QRS (especially ≥ 150 ms with LBBB), the greater the electrical delay CRT corrects and the better the response. This is why QRS width, not just EF, drives the indication tiers.

BOARD TRAP

WRONG: degree of EF reduction is the best predictor of who responds to CRT. Discriminator: across trials QRS duration/morphology predicts response better than EF — a patient with EF 34% and QRS 100 ms responds worse than one with EF 35% and LBBB 160 ms.

8. Reverse remodeling

WHAT YOU SEE

Conductor arrow reverse remodeling, up EF, up exercise

WHAT IT ENCODES

CRT drives reverse remodeling: coordinated contraction reduces wall stress and LV volumes, so over months EF rises, exercise capacity and NYHA class improve, functional MR falls, and mortality + HF hospitalizations decrease (COMPANION, CARE-HF). 'Super-responders' can normalize EF.

BOARD TRAP

WRONG: CRT only relieves symptoms and has no effect on survival. Discriminator: CARE-HF and COMPANION showed CRT reduces all-cause mortality, not just symptoms — reverse remodeling is a structural, prognosis-changing effect.

9. Functional MR improves

WHAT YOU SEE

Functional MR label left wing

WHAT IT ENCODES

CRT reduces functional (secondary) mitral regurgitation by resynchronizing the LV walls and papillary muscles, improving mitral leaflet coaptation and reducing the dyssynchrony-driven regurgitant volume. Less MR means lower LA pressures and less pulmonary congestion.

BOARD TRAP

WRONG: significant functional MR in a CRT candidate must be surgically repaired before considering CRT. Discriminator: functional MR is secondary to LV remodeling — resynchronization often improves it medically, so optimize GDMT + CRT before escalating to mitral intervention.

10. CRT-D vs CRT-P

WHAT YOU SEE

CRT-P and CRT-D boxes

WHAT IT ENCODES

CRT-D adds an ICD/defibrillator and is chosen for younger, ischemic, or higher-SCD-risk patients with good life expectancy. CRT-P is pacing only — preferred in older/frail patients, non-ischemic etiology, or limited life expectancy where defibrillator shocks add burden without benefit. The decision is a shared SCD-risk vs goals-of-care judgment.

BOARD TRAP

WRONG: every CRT patient should automatically receive a defibrillator (CRT-D). Discriminator: in elderly/frail or limited-prognosis patients, CRT-P avoids inappropriate shocks and lead complications; DANISH-type data show ICD benefit is attenuated in older non-ischemic patients.

11. Prevent pacing-induced CM

WHAT YOU SEE

Bottom-left: prevent pacing-induced cardiomyopathy

WHAT IT ENCODES

In a patient who needs frequent/chronic RV pacing AND has EF $\leq 35\%$ (or is expected to pace $>40\%$ of the time), use CRT (biventricular) rather than RV-only pacing to prevent pacing-induced cardiomyopathy. Chronic RV apical pacing creates an iatrogenic LBBB-like dyssynchrony (BLOCK-HF supports CRT here). Conduction-system pacing (His/left bundle branch area pacing) is an emerging alternative.

BOARD TRAP

WRONG: chronic RV apical pacing is harmless to the failing heart. Discriminator: high-burden RV pacing induces dyssynchrony and can drop EF (pacing-induced cardiomyopathy) — anticipate it and choose CRT or conduction-system pacing in low-EF patients needing a pacemaker.

12. 3-month GDMT wait

WHAT YOU SEE

3-month GDMT wait note

WHAT IT ENCODES

Wait ≥ 3 months on optimized GDMT before implanting CRT (and before primary-prevention ICD), because medical therapy frequently recovers EF and resolves the indication. The exception: a patient who already needs pacing (complete heart block, AV block) is paced/resynchronized without waiting.

BOARD TRAP

WRONG: implant CRT at the index admission in a newly diagnosed low-EF patient with LBBB. Discriminator: unless there's a bradycardia/heart-block pacing need, you give ≥ 3 months of GDMT first and re-image — premature device implant is a common board distractor.

13. Avoid CRT — narrow QRS / RBBB

WHAT YOU SEE

ECG ticker: RBBB X, narrow QRS X, won't work coffins

WHAT IT ENCODES

CRT will NOT work — and may harm — with a narrow QRS (<130 , certainly <120 ms) or with RBBB lacking LBBB-criteria delay, because there is no LV dyssynchrony to correct. Echo-CRT randomized narrow-QRS (<130 ms) HF patients and found CRT INCREASED mortality. Mechanical dyssynchrony on echo is NOT an indication when the QRS is narrow.

BOARD TRAP

WRONG: narrow-QRS HF with echo-demonstrated mechanical dyssynchrony benefits from CRT. Discriminator: Echo-CRT showed harm (increased mortality) when QRS <130 ms — CRT is an ELECTRICAL-delay therapy; never implant it for narrow QRS regardless of echo dyssynchrony.